

# RIJA FATIMA

AI Engineer | Intelligent Systems Developer

+92 300 1287034 | rijafatima59@gmail.com | Bahawalpur, Pakistan | [linkedin.com/in/rijafatima92](https://www.linkedin.com/in/rijafatima92) | [github.com/RijaFatima92](https://github.com/RijaFatima92)

## SUMMARY

---

Computer Science undergraduate at FAST NUCES with a focus on intelligent systems and algorithmic problem-solving. Comfortable taking a project from a rough idea to something that actually works — whether that's an AI agent competing in a real-time game, a full-stack web app, or a core data structure built to mimic a real product. Enjoys moving between algorithm design, backend logic, and interface work to ship something end-to-end.

## TECHNICAL SKILLS

---

**AI & Algorithmic Logic:** Autonomous Agents, Pathfinding (A\*, Greedy Best-First Search), Adversarial Search & Minimax with Alpha-Beta Pruning, Constraint Satisfaction Problems, Heuristic Optimization, Prompt Engineering

**Data Structures & Programming Fundamentals:** C++, Python, Hash Maps, Arrays, File Handling, Object-Oriented Programming

**Backend & Web Development:** Node.js, Express.js, REST API Design, PHP, MERN Stack, JWT Authentication, SQL

**Frontend & Interaction Design:** React.js, JavaScript, HTML5/CSS3, Figma (Auto Layout, Component Variants, Prototyping)

**Systems & Security:** Ubuntu/Linux, Shell Scripting, Video Steganography, DES Encryption, Security Analysis (XSS/CSRF)

## PROJECTS

---

**AI Autonomous Agent — Battle City** Python | Pygame | Heuristic & Adversarial Search

- Built an AI-controlled tank agent for a Battle City-style game that has to navigate a grid, dodge enemies, and react to changing threats in real time.
- Implemented and compared four search strategies — BFS, Greedy Best-First Search, A\*, and Minimax with Alpha-Beta Pruning — to see how each handled pathfinding and combat decisions differently.
- Worked with a partner to add a CSP-based map generator, so the game could produce new, solvable layouts instead of relying on one fixed map.

**Mini\_Instagram** C++ | Data Structures

- Recreated the core of Instagram — sign up, log in, post a (text-based) photo, and view a timeline — as a Data Structures course project.
- Used a hash map as the backend storage layer, designing it around how the app actually reads and writes data so lookups and inserts stayed fast as users and posts grew.

**Soul-Sisters Web App** MERN Stack (MongoDB, Express, React, Node.js) | REST APIs

- Built a full-stack platform end-to-end, from the database schema to the interface, to manage a product's complete lifecycle.
- Added JWT authentication and a modular REST API so the client and server could exchange data securely without unnecessary overhead.

**Video Steganography & Security System** Python | DES Encryption | Information Security

- Built a tool that hides DES-encrypted data inside video frames, so a payload can travel disguised as an ordinary video file.
- Stress-tested the system against basic attacks to see how well the hidden data held up, and documented what worked and what didn't.

**Connect\_four\_Game** C++ | Programming Fundamentals

- Built a two-player Connect Four game with a console-based interface as an early Programming Fundamentals course project.
- Used arrays to manage the game board and added file handling to save scores and game data between sessions.

**SkillBridge Interactive Platform** Figma | UI/UX Architecture | HCI

- Ran market-gap research to figure out what a responsive design system actually needed, then built it in Figma with Auto Layout and Component Variants.
- Delivered interactive, stakeholder-ready prototypes that stayed grounded in what was realistically feasible to implement.

## EDUCATION

---

**BS in Computer Science** 2023 – 2027

FAST NUCES, Chiniot-Faisalabad Campus

**Intermediate — pre-engineering** 2020 – 2022

Superior Group of Colleges, Bahawalpur